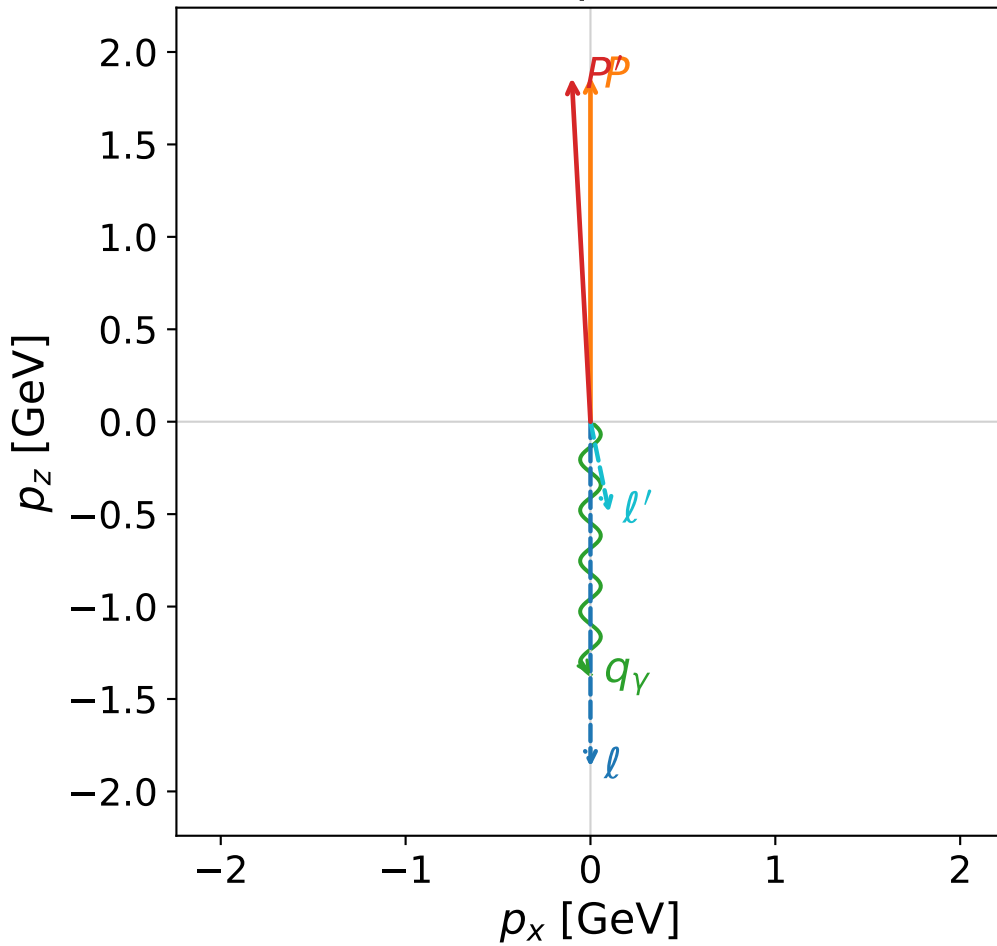
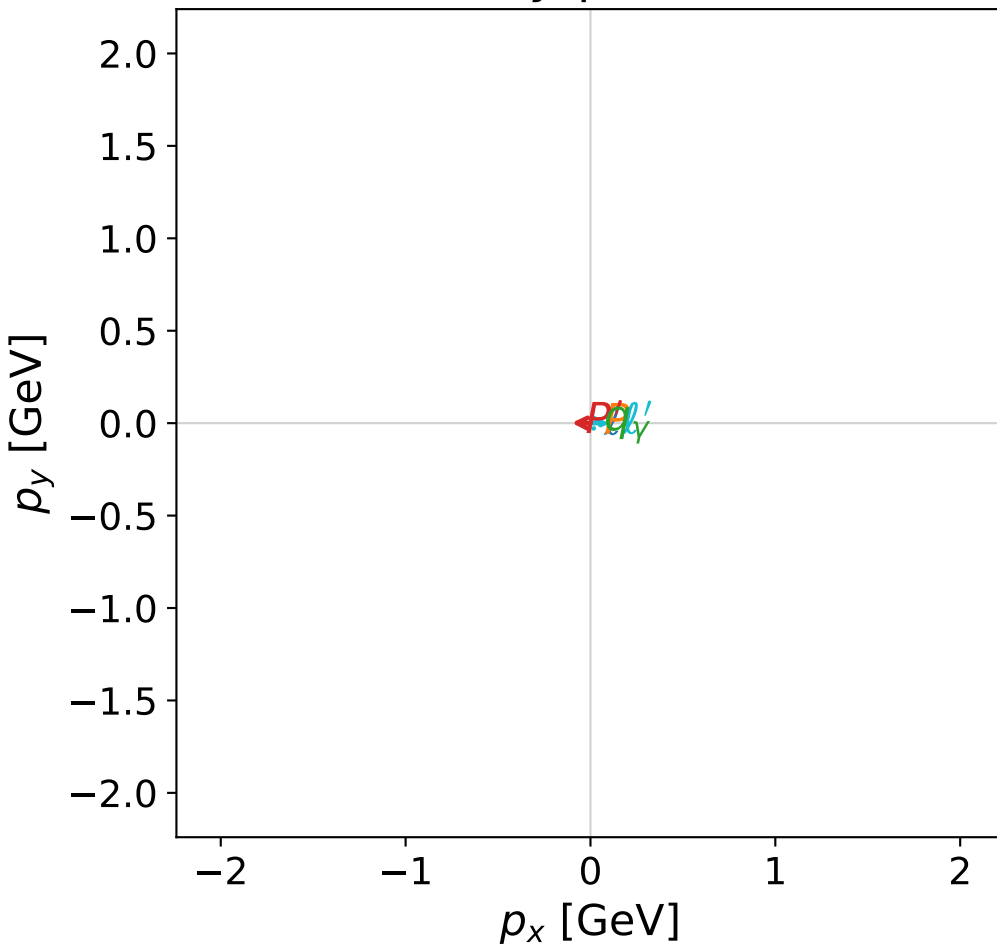


coherent mixing angles: maximum $C_{e\gamma}$ configuration (gradient_local_minimum)

z--x plane



x--y plane

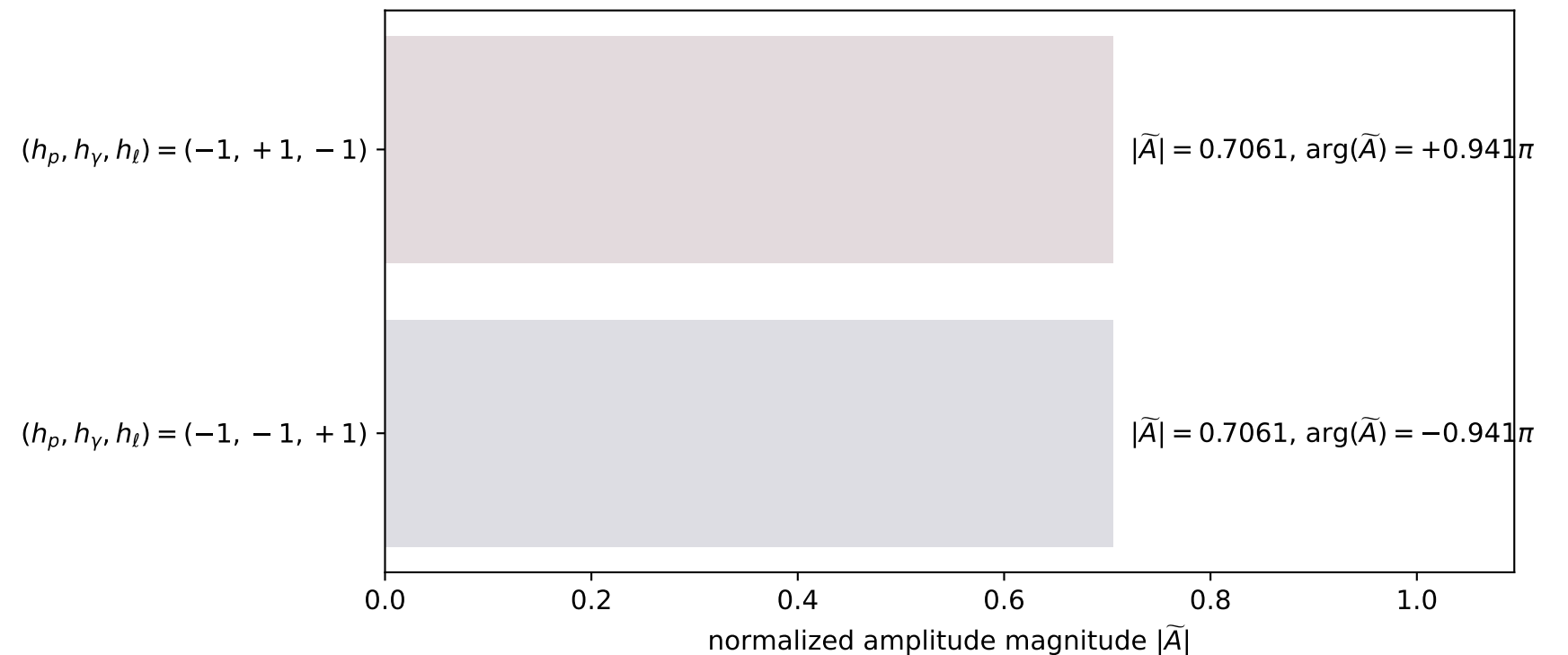


max_c_e_gamma_mixing_angles_polarization_03_local_minimum_3
 $C_{\{e\gamma\}}=1$
 region: polarization_cluster_03_local_minimum_3
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.354415$ rad
 incoming lepton: $-0.705847|+\rangle +0.708364|-\rangle$
 proton mixing angle: $\alpha_p=1.6068304$ rad
 incoming proton: $-0.0360263|+\rangle +0.999351|-\rangle$

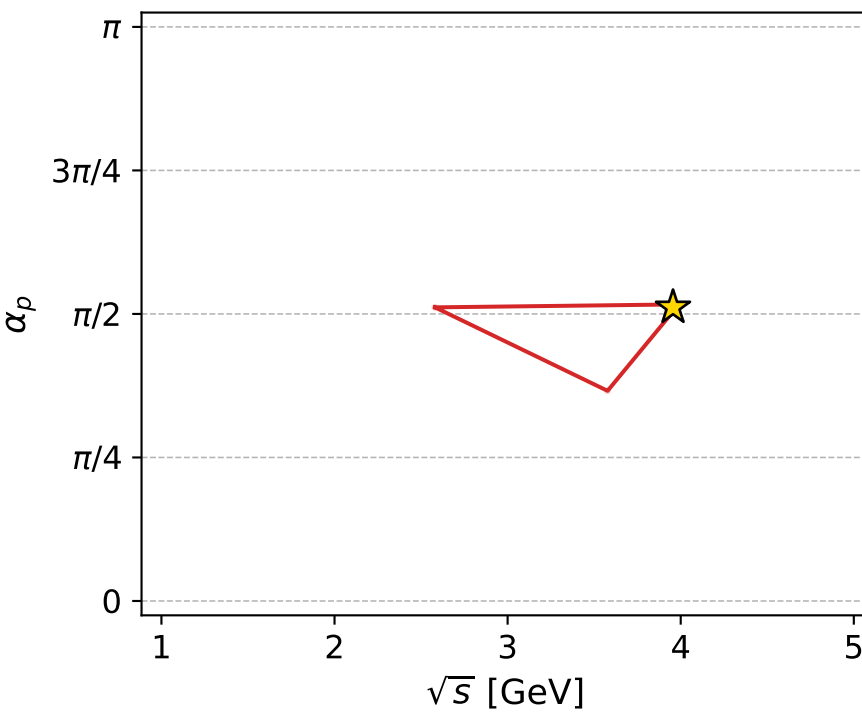
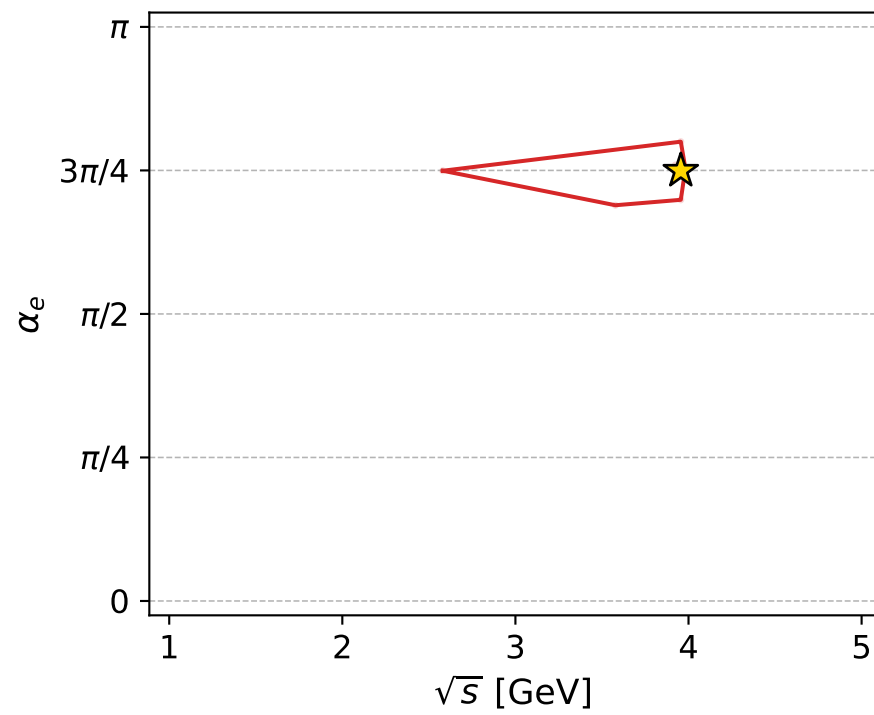
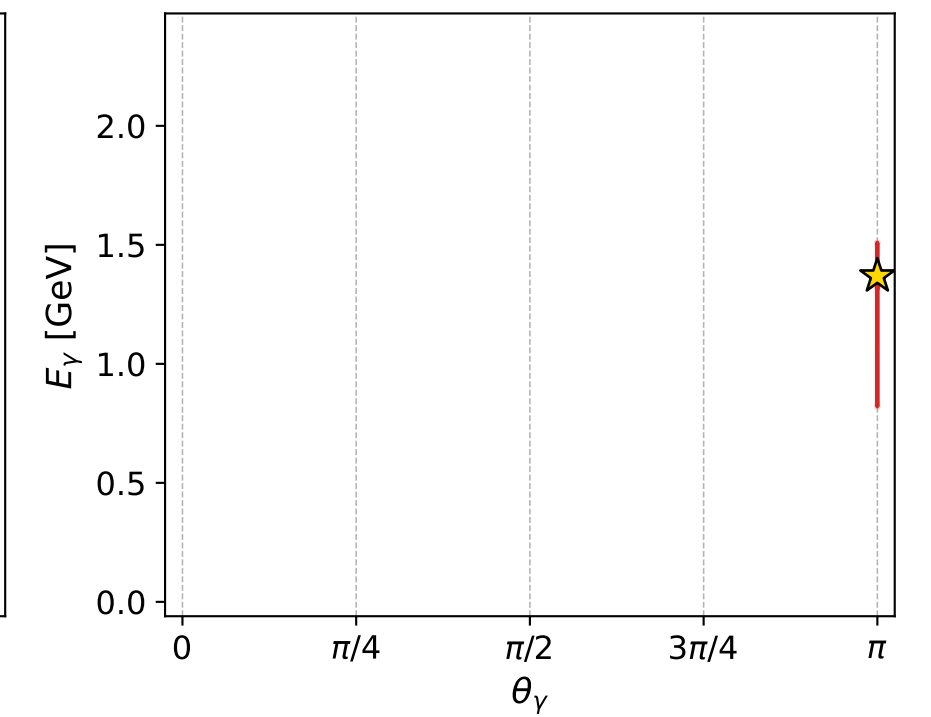
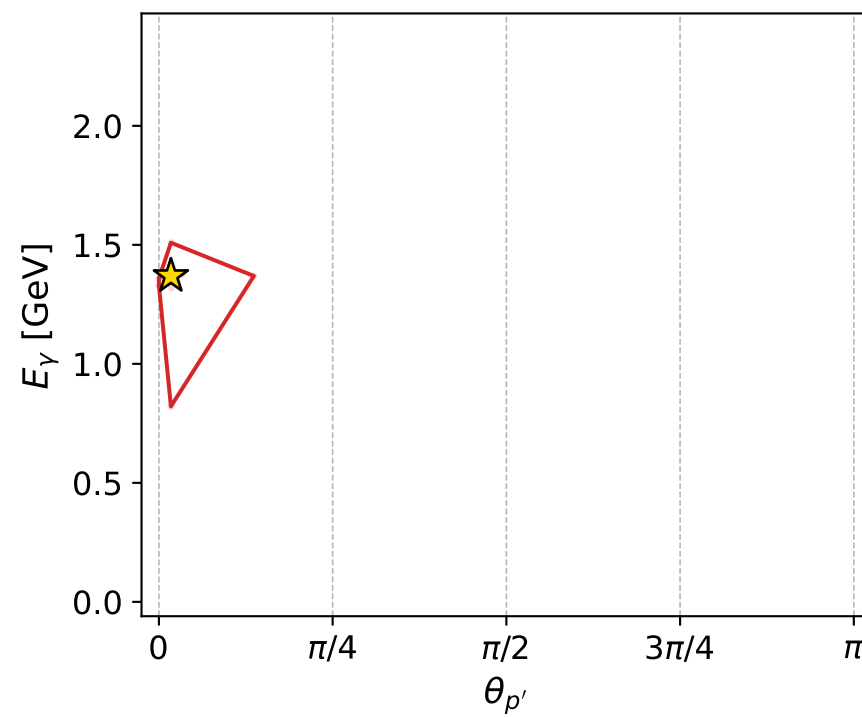
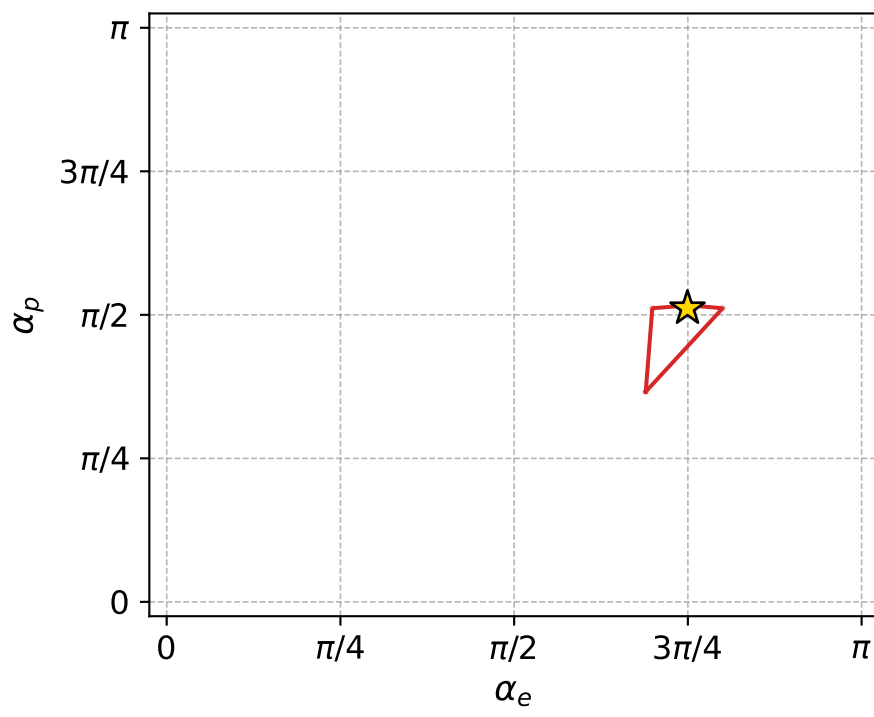
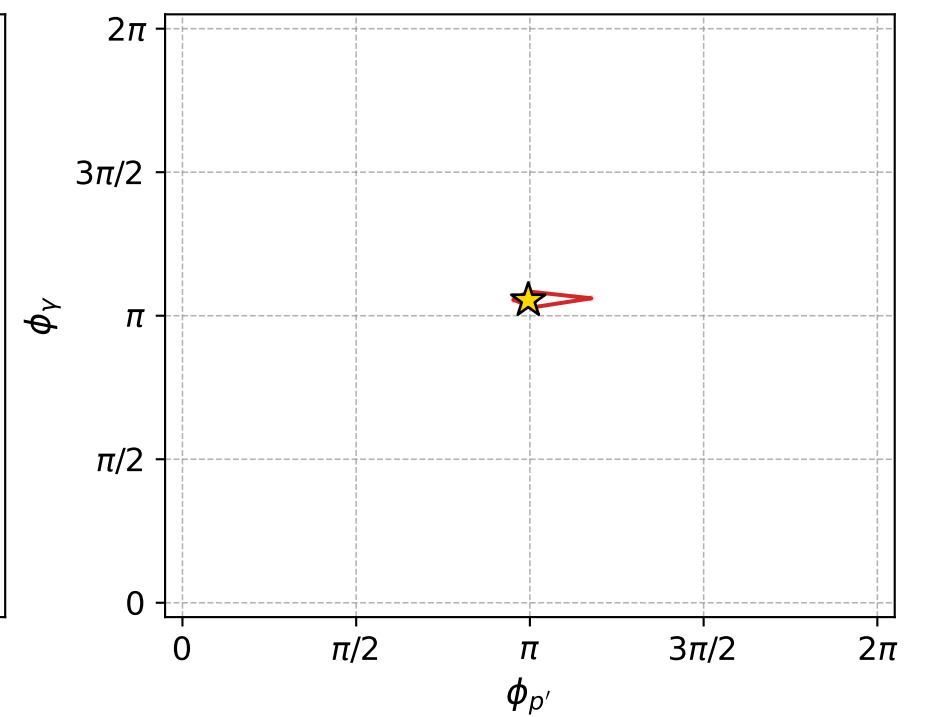
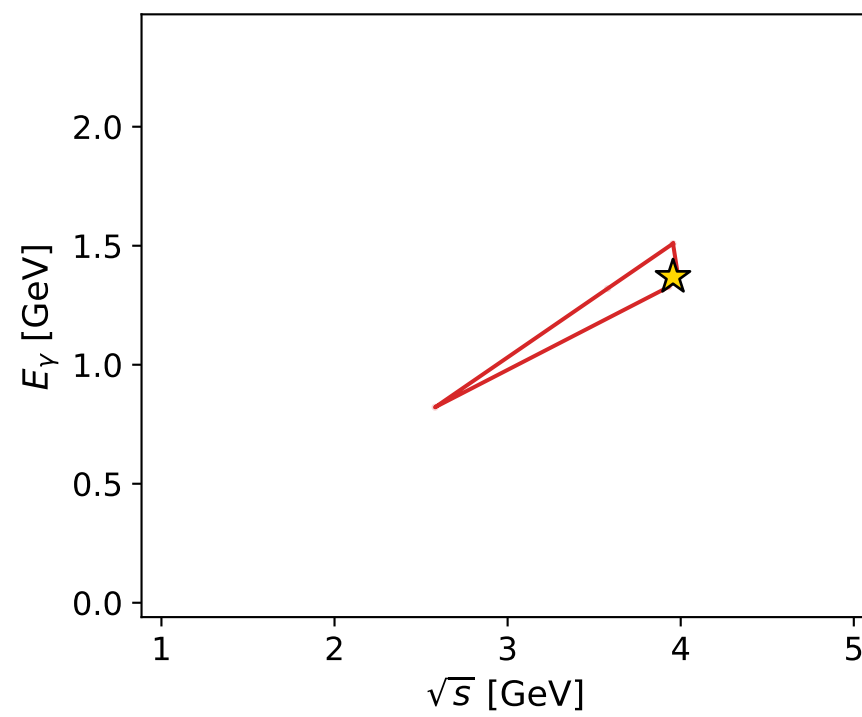
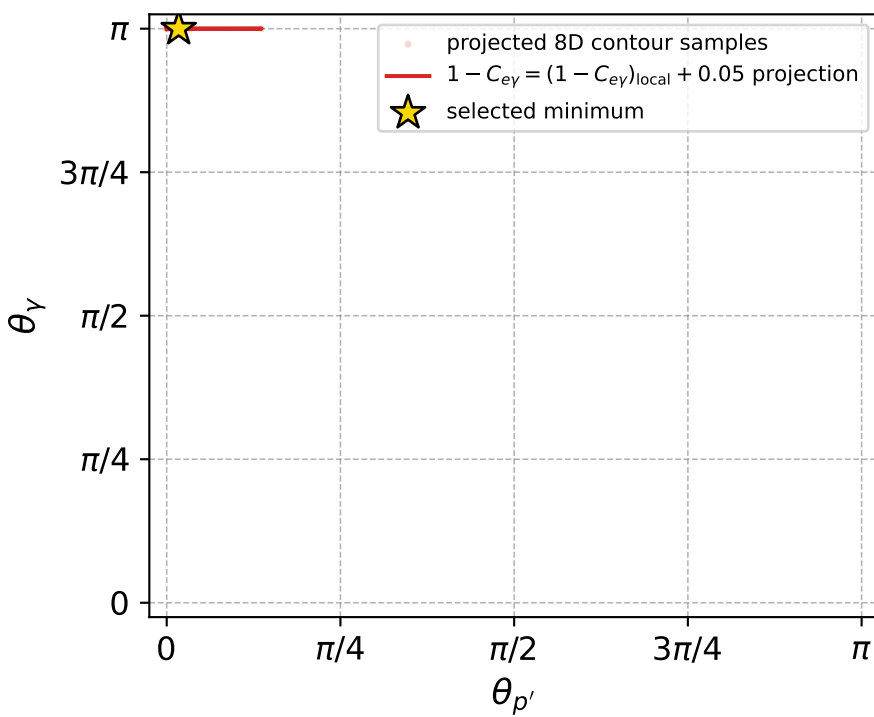
$s=15.646$, $\sqrt{s}=3.9555$, $|\vec{P}|=1.86653$, $|\vec{P}'|=1.86254$
 $\theta_{p'}=0.0543166$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.1268$, $\phi_\gamma=3.31099$
 $E_\gamma=1.36911$, $\theta_{l'}=2.93836$, $\phi_{l'}=6.26839$
 $Q^2=0.0384902$, $x_B=0.00355034$, $t=-0.0102573$, $W^2=11.6826$, $y=0.734199$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8665$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8665$
 P $E= 2.0890$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8665$
 ℓ' $E= 0.5010$ $p_x= 0.1011$ $p_y= -0.0015$ $p_z= -0.4907$
 P' $E= 2.0854$ $p_x= -0.1011$ $p_y= 0.0015$ $p_z= 1.8598$
 q_γ $E= 1.3691$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.3691$

Leading normalized final-state amplitudes
 (N=2, retained=99.7%)



electron: polarization cluster P3, local minimum 3; pairwise projections of the 8D $1 - C_{e\gamma} = (1 - C_{e\gamma})_{\text{local}} + 0.05$ contour



selected local minimum

$\text{one_minus_C_e_gamma}(\text{local}) = 1.1827961\text{e-}10$
 $\text{one_minus_C_e_gamma contour} = 0.05$
 $\text{above global minimum} = 7.502543\text{e-}11$
 $\text{8D contour samples} = 777$
 $\text{configured 8D directions} = 1536$

selected phase-space point:

$\text{sqrt_s} = 3.955497$
 $\text{theta_p_out} = 0.05431658$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.369109$
 $\text{phi_p_out} = 3.1268$
 $\text{phi_gamma_out} = 3.310992$
 $\text{alpha_e} = 2.354415$
 $\text{alpha_p} = 1.60683$